VCF 9.x triage cheat sheet

Work down the steps in order. Each one has a command, what a good result looks like, and what to do when it is not good. Commands are read-only unless marked. Replace `<…>` with your values.

Which layer owns it?

	Management component	Core infrastructure
What	VCF Operations, VCF Automation, identity broker, Ops for networks, log management, services runtime	vCenter, ESX, NSX Managers and Edges, vSAN, SDDC Manager
Tasks	Build > Lifecycle > VCF Management > Tasks	SDDC Manager Tasks, Build > Lifecycle > VCF Instances
Logs	Runtime pods in `vcf-fleet-lcm`, `vcf-sddc-lcm`	`/var/log/vmware/vcf/` on SDDC Manager

The steps

1. **Capture the IDs**
   Task details: SDDC Manager Tasks, or VCF Operations Build > Lifecycle > VCF Management > Tasks (9.1)

```bash
date -u    # keep the time with its offset, and check the offset each log file shows
```

   GOOD The ticket has the task ID, the reference ID or token, the failed subtask, and the time with its UTC offset.
   IF NOT Do not retry yet. A retry creates a new task and the first failure becomes harder to find.

2. **Read the cause**

```bash
grep -Ril "<reference-id>" /var/log/vmware/vcf/
grep -n -B 20 -A 60 "<reference-id>" /var/log/vmware/vcf/<service>/<service>.log | less
```

   GOOD The last "Caused by" names something concrete: a host, an account, a certificate, a file system.
   IF NOT Search by the minute instead of the ID, in the offset that log uses. For a management component, read the runtime pod logs (toolkit below).

3. **Check for locks and stale credentials**

```bash
/opt/vmware/sddc-support/sos --health-check --domain-name <domain>
```

   GOOD A clean run. On 5.x and 9.0.x a stale lock fails this check even with every task complete. On 9.0, so can "Failed to get details" against an ESXi root account.
   IF NOT The log line names the lock holder. In progress and really hung: cancel it (API below). Failed: fix the cause and retry, because cancel returns 409. Nothing to cancel: a lock-row KB, with a snapshot. "Failed to get details": remediate with the password that works on the component today.

4. **Check DNS**

```bash
ip=$(dig +short <fqdn> | tail -1); dig +short -x "$ip"
```

   GOOD The reverse record returns the same FQDN, in lowercase. Run it from the component that failed.
   IF NOT Fix the record or its case, wait for the TTL, then Retry Precheck.

5. **Check time**

```bash
chronyc tracking 2>/dev/null | grep -E 'System time|Leap' || ntpq -p
date -u
```

   GOOD Every node uses the same source and shows the same UTC time to within a second or two.
   IF NOT Fix NTP first: on 9.0 embedded SSO, drift shows as "JWT token is expired" (KB 409353). After an NTP change, confirm the setting on the appliance itself.

6. **Check disk**

```bash
# SDDC Manager
df -h /data /opt /var/log 2>/dev/null
# vCenter
df -h /storage/log /storage/seat /storage/db /storage/core
```

   GOOD SDDC Manager: more than 10 GB free in /opt and more than 1 GB for the database (the precheck's critical limits). vCenter: no partition close to full.
   IF NOT Collect a support bundle before you clean anything. Then use the documented clean-up, not rm: on 9.x, DELETE DOWNLOAD under Binary Management, then bundle_cleanup.py if that fails.

7. **Check the path**

```bash
curl -skv --max-time 10 https://<target>/ -o /dev/null 2>&1 | grep -E "Connected|SSL|HTTP/"
```

GOOD Connected, a TLS handshake, an HTTP status. Test from the address that makes the call: in 9.1, often a runtime node.
IF NOT Timeout: firewall or route. TLS error: inspection proxy or untrusted CA. 403: allowlist or token.

8. **Retry the smallest thing**

Retry Precheck, or Retry Task. Broadcom doesn't document where a manual retry restarts, so check what the failed step left behind. These commands change state.

```bash
# TOKEN: accessToken from POST https://localhost/v1/tokens (SDDC Manager admin)
# Retry a failed task
curl -sk -X PATCH -H "Authorization: Bearer $TOKEN" https://localhost/v1/tasks/<id>
# Cancel a task still in progress that will not complete (a failed one returns 409)
curl -sk -X DELETE -H "Authorization: Bearer $TOKEN" https://localhost/v1/tasks/<id>
# Cancel a failed password task
curl -sk -X DELETE -H "Authorization: Bearer $TOKEN" \
  https://localhost/v1/credentials/tasks/<id>
```

GOOD The task passes the step that failed.
IF NOT Same error again: back to step 2, the cause is not fixed. Password tasks stay on the credentials path, never the generic API.

Symptom, first check, next action

You see	Check	Then
UnknownHostException, installer DNS validation, 502 in the installer	`dig +short <fqdn>; dig +short -x <ip>`	Add or fix the record in lowercase, then Retry Precheck.
"JWT token is expired" on 9.0 embedded SSO	`date -u on VCF Operations and on vCenter`	Clocks differ: fix the time source, then retry (KB 409353).
"Resource is already locked", "Error in acquiring resource locks"	`The log names the holder; GET /v1/tasks/<owner>`	In progress and hung: cancel it. Failed: fix the cause and retry. Nothing to cancel: delete a lock row only with a snapshot and the lock KB.
Account locked, InvalidLogin, 403 in operationsmanager.log	`sos --health-check on 9.0 (Failed to get details)`	Unlocking first fails while a service still holds the old password. For the NSX service account: add SDDC Manager to the lockout-immune list, rotate it, verify, then rotate the rest (KB 449684).
PKIX path building failed, certificate_unknown(46)	`openssl s_client -connect <host>:443 </dev/null | openssl x509 -noout -issuer -enddate`	Expired: replace. Unknown issuer: TLS inspection on the path, exempt it, or import the CA.
Signed certificate rejected	`openssl x509 -in cert.crt -pubkey -noout | sha256sum; openssl req -in csr.csr -pubkey -noout | sha256sum`	Hashes differ: the certificate was signed from another CSR. Sign the current one.
"no healthy upstream" on vCenter	`df -h; service-control --status --all; the VECS expiry loop`	Free the full partition or replace the expired certificate, then start services. A reboot alone frees no partition and replaces no certificate.
9.1 runtime tasks stuck	`kubectl get pods -A | grep -vE 'Running|Completed'`	Fix pod health first. Restart a lifecycle service with lcm_service_support.sh, and delete pods only where a KB names them.
Depot 403, "Bundle not found", download stalls	`curl -I https://eapi.broadcom.com from a runtime node`	Allow every runtime node, not only SDDC Manager. Check the token or activation code, and TLS inspection.
NSX host preparation or TEP failures	`vmkping -S vxlan -s 1572 -I vmk10 -d <edge-tep>`	Fails with -d: MTU on the path is under 1600. Fix the network, then Resolve in NSX.
VCF Operations slow, node offline, cluster not ready	`curl -sk https://<node>/casa/cluster/status; df -h /storage/db /storage/log`	One node must be MASTER. Add disk to every node, identically sized. Never revert to a snapshot.

Log map

SDDC Manager	`/var/log/vmware/vcf/` : `lcm/lcm-debug.log` (prechecks, bundles, upgrades), `domainmanager/` (domains, clusters, hosts), `operationsmanager/` (passwords, certificates, backups, commissioning), `commonsvcs/` (tokens, SSH keys, API)
VCF Operations	`/storage/log/vcops/log/` : `collector.log` , `web-vcf.log` , `adapters/<Kind>/` , `casa/casa.log`

Collect before you escalate

```bash
cd /opt/vmware/sddc-support
# One workflow failed
./sos --sddc-manager-logs --zip
# The failure spans hosts, vCenter or NSX
./sos --collect-all-logs --zip --domain-name <domain>
```

Runtime (9.1)	Fleet: `vcf-fleet-lcm`. SDDC: `vcf-sddc-lcm`. Depot: `vcf-fleet-depot`. Platform: `vmsp-platform`
vCenter	`/var/log/vmware/`, `vsphere-ui/logs/`, `cis-license/license.log`, `vmware-updatemgr/`
NSX	`/var/log/cm-inventory/cm-inventory.log`, `/var/log/syslog`

```
  # 9.1 management services as well
  ./sos --sddc-manager-logs --vcf-mgmt-svcs-logs
  --zip
```

- **VCF Operations:** Control Panel > Support Bundles.
- **Runtime down, fleet lifecycle unreachable:** the standalone bundle script (KB 444466).
- **UI problems:** a browser HAR file.
- **Always:** the IDs from step 1, the time with its UTC offset, and every action taken since.

Rules

One operation at a time on a scope. Health before retry, retry before rebuild. Bundle before clean-up. Snapshot before any database change, and make database changes only when a KB for your exact build says to.

Services runtime toolkit (9.1, read-only)

bash

```
# Control-plane node IP: Build > Lifecycle > VCF Management > Components > VCF Services Runtime > Nodes
ssh vmware-system-user@<control-plane-node>
sudo -i
export KUBECONFIG=/etc/kubernetes/admin.conf
kubectl get nodes -o wide
kubectl get pods -A | grep -vE 'Running|Completed'
kubectl get events -n vcf-fleet-lcm --sort-by=.metadata.creationTimestamp | tail -30
kubectl get pods -n vcf-fleet-lcm   # copy a fleet build pod's name
kubectl logs <pod> -n vcf-fleet-lcm --tail=200
kubectl -n vcf-fleet-lcm exec vcf-fleet-lcm-db-0 -- patronictl list  # one leader, no lag

# Changes, on SDDC Manager. Stuck fleet or SDDC lifecycle task (KB 435569): report, then restart one
service
lcm_service_support.sh -u <admin_user> -s <runtime-fqdn> -t <task_id>
lcm_service_support.sh -u <admin_user> -s <runtime-fqdn> -r fleetbuild    # or fleetupgrade, sddcbuild,
sddcupgrade
```

Find `lcm_service_support.sh` and run it as KB 435569 describes for your build. Don't delete pods or `vcf-svc-*` VMs to restart them, unless a KB names them for your symptom, as KB 443718 does. More in the services runtime for VI admins.

Not official vendor guidance. Confirm against current Broadcom documentation for your build before changing production.